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***B.Tech. Degree V Semester Supplementary Examination
November 2020/April 2021***

**ME 15-1504 THERMAL ENGINEERING
(2015 Scheme)**

Time: 3 Hours

Maximum Marks: 60

**PART A
(Answer ALL questions)**

(10 × 2 = 20)

- I. (a) Explain Higher and Lower Calorific values.
- (b) Explain with a neat sketch the orsat apparatus.
- (c) Obtain the expression of efficiency of diesel cycle in terms of compression ratio r and ratio of specific heats γ and cutoff ratio ρ .
- (d) Explain actual port timing diagram of a 2-S petrol engine. Explain scavenging using it.
- (e) Explain retardation on an IC engine. Explain if this test can be conducted in all engines.
- (f) What is supercharging? What are its objectives?
- (g) Explain with a neat sketch magneto ignition system of a 6 cylinder engine.
- (h) Explain free piston engine with neat sketches.
- (i) Explain post combustion and knock in SI engines.
- (j) Explain dissociation and its effect in IC engines.



PART B

(4 × 10 = 40)

- II. (a) Explain Adiabatic flame temperature. (2)
 - (b) A sample of fuel has the following percentage composition. (8)
Carbon-86%, Hydrogen-8%, Sulphur- 3%, Oxygen 1% and ash 2%.
For an airfuel ratio of 12:1, calculate:
(i) The mixture strength as percentage rich or weak.
(ii) Volumetric analysis of dry product of combustion.
- OR**
- III. (a) Explain Bomb Calorimeter with neat sketch. (2)
 - (b) The higher heating value of kerosene at constant volume whose ultimate analysis is 83% Carbon and 17% Hydrogen was found to be 45670 kJ/kg. Calculate the other three heating values. (8)
- IV. (a) Derive the expression of air standard efficiency of an otto cycle in terms of compression ratio, and ratio of specific heats. Also write the expression for and mean effective pressure. (2)
 - (b) The volume ratios of compression and expansion for a diesel engine as measured from an indicator diagram are 15.3 and 7.5 respectively. The pressure and temperature at the beginning of compression are 1 bar and 29°C. Assuming an ideal engine, determine the mean effective pressure, ratio of maximum pressure to mean effective pressure and cycle efficiency. Also find the fuel consumption per kWh if the indicated thermal efficiency is 0.5 of ideal efficiency, Mechanical efficiency is 0.8 and the Calorific value of oil is 42000 kJ/kg. (8)

OR

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- V. (a) Explain Willian's line method of testing in IC engines. What are the limitations of this test? (2)
- (b) The following observations were recorded in a test of 1 hour duration on a single cylinder oil engine working on four stroke cycle. (8)
- | | |
|---------------------------|---------------|
| Bore | = 300 mm |
| Stroke | = 450 mm |
| Fuel used | = 8.8 kg |
| Calorific value of fuel | = 41800 kJ/kg |
| Average speed | = 200 rpm |
| mep | = 5.8 bar |
| Brake friction load | = 1860 N |
| Quantity of cooling water | = 650 kg |
| Temperature rise | = 22°C |
| Diameter of the wheel | = 1.22 m |
- Calculate:
- (i) Mechanical efficiency.
- (ii) Brake thermal efficiency and draw the heat balance sheet.
- VI. (a) Correlate the actual valve timing diagram and indicator diagram of a 4S SI engine. (5)
- (b) With neat sketch explain the modifications required in the battery ignition system of a 4 Cylinder SI engine to convert it in to the magneto ignition system of a 3 cylinder SI engine. (5)
- OR**
- VII. (a) Explain Mist lubrication system and its advantages and disadvantages. (5)
- (b) Explain Stratified charged engine and wankal engine. (5)
- VIII. (a) Explain Effects and control of Detonation. (5)
- (b) Explain Induction swirl and induction ports in SI engines. (5)
- OR**
- IX. (a) Explain Combustion phenomenon in CI engines with Pressure, Crank angle diagram. (5)
- (b) Explain Types of combustion chambers in CI engines. (5)
